



Dataset Status & Source

Primary Dataset

Custom self-collected dataset — capture parameters defined: 30–50% horizontal overlap, fixed camera height, consistent lighting across 5 distinct campus scene locations.

Secondary Dataset

Adobe Panoramas Dataset (downloaded) — reserved for baseline benchmarking and comparative evaluation.

Status

Primary: In progress (collection planned). Secondary: Ready for baseline evaluation.

Image Preprocessing Plan



01

Color Space Conversion

Convert RGB $\rightarrow$ Grayscale.
ORB/SIFT operate on intensity gradients; color not required for detection.

02

Resizing

Downscale high-res images to a max width of 1024px to enable real-time development speeds (memory & compute savings).

03

Distortion Correction (Optional)

Evaluate lens distortion calibration after initial tests; apply if residual geometric error remains significant.

Baseline Methodology (ORB)



Feature Detection

Use ORB (Oriented FAST + Rotated BRIEF) for efficient keypoint detection and binary descriptors.



Feature Matching

Brute-Force matcher with Hamming distance to match binary descriptors.



Geometric Transform

Compute Homography matrix from top 100 matched keypoints (baseline: no advanced outlier rejection).



Stitching

Simple perspective warping and direct overlay (pixel overwrite) to form baseline panorama.



Initial Implementation Results



Result:

Baseline detects and matches features and computes a rough homography to produce a preliminary stitch (visible seam and rough alignment remain).



Technical Challenges & Solutions

1

Challenge 1 — High-Res Noise

12MP phone images cause ORB to detect many ambiguous features, slowing processing.

2

Solution

Aggressively resize to fixed height (600px) before processing to remove high-frequency noise and reduce keypoint clutter.

3

Challenge 2 — False Positive Matches

Full-image matching produced incorrect matches from background/non-overlap areas, creating spurious correspondences ("crazy lines").

4

Solution (ROI-Based Matching)

Crop expected overlap regions (Right 40% of Image 1, Left 40% of Image 2), run ORB inside ROIs, then shift coordinates back to global space for Homography. Eliminated false matches.

5

Next Steps (M3)

Implement multi-band blending/feathering to remove visible seam and add an iterative stitching loop for 3+ images.